The second Kahn–Kalai conjecture

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

For every finite simple graph H with $h \geq 1$ edges and at most n vertices, the threshold for $G(n,p)$ to contain a copy of H is at most $C\, p_{\mathrm{E}}(n,H)(1+\log_2 h)$ with a universal constant C, where $p_{\mathrm{E}}(n,H)$ is the least density at which every subgraph of H has expected copy count at least one half. This is the second Kahn–Kalai conjecture. The proof given here is self-contained and uses only elementary probability.

1 The result and an overview of the proof

1.1 Statement

All graphs are finite and simple. A *copy* of F in G is a subgraph of G isomorphic to F, where a subgraph consists of a set of vertices of G and a set of edges of G among them. Copies need not be induced. Write $N(G,F)$ for the number of copies of F in G. Let $G(n,p)$ be the random graph on $[n]=\{1,\dots,n\}$ in which each of the $\binom{n}{2}$ possible edges is present independently with probability p. For a graph H with at least one edge and at most n vertices, define

$$p_{\mathrm{c}}(n,H) = \inf\{p\in[0,1] : \mathbb{P}(N(G(n,p),H)>0) \geq \tfrac{1}{2}\},$$
$$p_{\mathrm{E}}(n,H) = \inf\{p\in[0,1] : \mathbb{E}N(G(n,p),F) \geq \tfrac{1}{2} \text{ for every subgraph } F\subseteq H\}.$$

The first is the *containment threshold*, the second the *graph expectation threshold*. Four elementary facts will be used.

(a) Each copy of F in K_n is present in $G(n,p)$ with probability $p^{e(F)}$, so

$$\mathbb{E}N(G(n,p),F) = N(K_n,F)\,p^{e(F)}. \tag{1.1}$$

(b) *Both infima are attained.* By (1.1) each constraint in p_{E} defines a closed subset of $[0,1]$. It contains $p=1$ because $N(K_n,F)\geq 1$ when F has at most n vertices. There are finitely many F. Likewise $\mathbb{P}(N(G(n,p),H)>0)$ is a polynomial in p with value 1 at $p=1$.

(c) $p_{\mathrm{E}}(n,H)>0$. For F a single edge the constraint is $\binom{n}{2}p\geq\frac{1}{2}$.

(d) $p_{\mathrm{E}}(n,H)\leq p_{\mathrm{c}}(n,H)$. If H appears with probability at least $\frac{1}{2}$, so does every $F\subseteq H$, and $\mathbb{E}N\geq\mathbb{P}(N>0)$.

Theorem 1.1. *For every $n\geq 2$ and every finite simple graph H with $h=e(H)\geq 1$ edges and at most n vertices,*

$$p_{\mathrm{c}}(n,H) \leq \min\{1,\ 2048\mathrm{e}^{50} p_{\mathrm{E}}(n,H)(1+\log_2 h)\}. \tag{1.2}$$

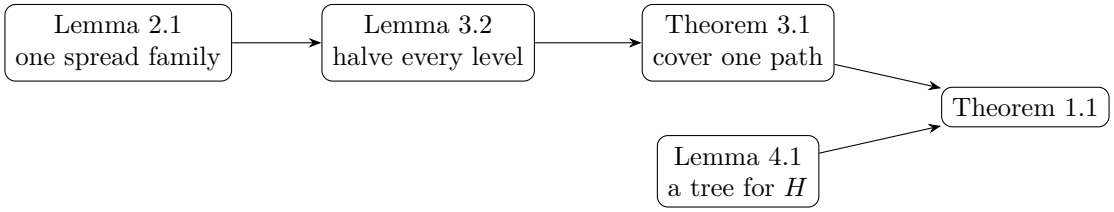
In particular,

$$p_{\mathrm{c}}(n,H) \leq 6144\mathrm{e}^{50} p_{\mathrm{E}}(n,H)\log_2 n.$$

Here e is the base of the natural logarithm. The constants are universal, and H may vary with n. Kahn and Kalai use expected count one instead of one half. By (1.1) the constraint of a subgraph with $x \geq 1$ edges then changes by the factor $2^{1/x} \leq 2$, so the expectation threshold changes by a factor between 1 and 2 and the conjecture is the same.

1.2 Proof at a glance

A probability law on subsets A of a finite set X is *a-spread* if $\mathbb{P}(J \subseteq A) \leq a^{|J|}$ for every nonempty $J \subseteq X$. The proof has three layers.



Layer 3: graphs give a tree (Lemma 4.1). Put $q = p_{\mathrm{E}}(n, H)$. One builds a chain of edge sets $\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = H$ and a rooted tree. A node at depth i is a copy of H_i on $[n]$, together with the copies of $H_1, \ldots, H_{i-1}$ through which it was reached. Its children are all copies of H_{i+1} containing it, chosen uniformly. The arc to a child is labelled by the set of added edges. The chain is chosen by a maximisation, and this has two consequences. First, the first-moment constraints at density q force $|H_{i-1}| < |H_i|/17$, so label sizes grow by a factor of at least 16 per level. Second, maximality makes every child law $128q$-spread. Labels along a root-to-leaf path are disjoint and their union is a copy of H.

Layer 2: covering a path of a tree (Theorem 3.1, Lemma 3.2). Consider any tree with labels disjoint along paths, σ-spread child laws and label sizes growing by a factor of 16. Then a random set of density $16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k)$ contains all labels of some root-to-leaf path with probability at least $\frac{2}{3}$. Here ℓ_k bounds the largest label size. The proof iterates a *reduction*. One random set W of density $\rho = 4\mathrm{e}^{50}\sigma$ is used at every node at once. With probability at least $\frac{3}{4}$ it yields a pruned, reweighted tree whose labels are the uncovered parts $A \setminus W$ of old labels. Every level's size bound is halved and the spread parameter worsens by a factor close to one. After $\lfloor \log_2 \ell_k \rfloor + 1$ successful reductions all labels are empty, so the union of the sets W used covers a path.

Layer 1: one family (Lemma 2.1). For a single a-spread family with sets of size at most m and a random set W of density $\rho \geq \mathrm{e}^{50}a$, a randomised rule picks a member A_b (the "target"). Two estimates hold. With probability at least $1 - \mathrm{e}^{-9m}$ the uncovered part $A_b \setminus W$ has size at most $m/2$. And conditioning on the target being b distorts the law of W *outside* A_b by a factor of at most e^{11m}.

The key point. Reusing the same W at all levels creates dependence between a node's choice and what happens below it. Disjointness along paths controls this. Everything below an arc with label A depends only on W outside A, which is exactly where the distortion estimate of Layer 1 applies. The distortion cost e^{12m_j} at level j is absorbed by the gain $\mathrm{e}^{-8m_{j+1}}$ at the next level, because $m_{j+1} \geq 16m_j$.

Where the constant comes from. $2048\mathrm{e}^{50} = \mathrm{e}^{50} \cdot 4 \cdot 4 \cdot 128$. The factor e^{50} is the ratio ρ/a in Lemma 2.1. The first 4 bounds the total worsening of spread over all reductions. The second 4 is the number of random sets drawn per needed success. The factor 128 is the ratio σ/q in Lemma 4.1.

1.3 Context and attribution

This subsection condenses the source's account of the literature. None of it is used in the proof, and the cited statements were not rechecked for this rewrite. Symbols introduced here ($q(\cdot)$, q_f, ℓ, L) are local to this subsection.

The conjecture. Kahn and Kalai [8] conjectured that the first-moment obstruction p_{E} determines p_{c} within a logarithmic factor, uniformly over H. For a fixed target, thresholds belong to the classical theory of Erdős and Rényi [5] and Bollobás [2]. Here the target may grow with n. A logarithm cannot be avoided. For a perfect matching on an even number n of vertices, $p_{\mathrm{E}} = \Theta(1/n)$ whereas $p_{\mathrm{c}} = \Theta(\log n/n)$ [6, 8], since isolated vertices of the host must disappear. Hamilton cycles show the same gap [8]. Other spanning targets have required different methods. For example, Riordan [14] proved that for fixed $p > 1/4$, $G(2^d, p)$ contains the d-dimensional cube with probability tending to one.

Why "second". Let $\mathcal{F} \subseteq 2^X$ be a nonempty proper increasing family on a finite set X. A *cover* is a family $\mathcal{G} \subseteq 2^X$ such that every $F \in \mathcal{F}$ contains some $S \in \mathcal{G}$. Its cost at density p is $\sum_{S\in\mathcal{G}} p^{|S|}$. The integral expectation threshold $q(\mathcal{F})$ is the largest p at which some cover has cost at most $\frac{1}{2}$. The fractional threshold $q_f(\mathcal{F})$ allows weights $w_S \geq 0$ with $\sum_{S\subseteq F} w_S \geq 1$ for all $F \in \mathcal{F}$ and cost $\sum_S w_S p^{|S|}$, with the same budget. Then $q(\mathcal{F}) \leq q_f(\mathcal{F}) \leq p_{\mathrm{c}}(\mathcal{F})$, where $p_{\mathrm{c}}(\mathcal{F})$ is the density at which a random subset lies in $\mathcal{F}$ with probability $\frac{1}{2}$. Talagrand [15, Section 6] introduced the fractional formulation and its dual description by spread measures. Alweiss, Lovett, Wu and Zhang [1], building on Lovett, Solomon and Zhang [9], developed the fragment method. Frankston, Kahn, Narayanan and Park [7] proved $p_{\mathrm{c}}(\mathcal{F}) = O(q_f(\mathcal{F}) \log \ell)$, where ℓ is the maximum of two and the largest size of a minimal member of $\mathcal{F}$. Park and Pham [13] proved the same with q in place of q_f, which is the abstract Kahn–Kalai conjecture. The two thresholds agree up to a universal constant factor [12, Theorem 1.1].

For the family $\mathcal{F}_H$ of edge sets containing a copy of H, the copies of any one subgraph form a cover. Hence $p_{\mathrm{E}}(n, H) \leq q(\mathcal{F}_H)$, and the abstract theorem does not bound p_{c} by p_{E}. Mossel, Niles-Weed, Sun and Zadik [11, Theorem 1] obtained a single logarithm for the modified threshold $\widetilde{p}_{\mathrm{E}}(n, H) = \min\{p : \mathbb{E}N(G(n,p), F) \geq N(H,F)/2 \text{ for every } F \subseteq H\}$, which equals $q_f(\mathcal{F}_H)$ [3, Equation (1)]. Dubroff, Kahn and Park [3, Theorems 1.2–1.5] proved $q_f(\mathcal{F}_H) = O(p_{\mathrm{E}}(n, H) \log^2 n)$, hence $p_{\mathrm{c}}(n, H) = O(p_{\mathrm{E}}(n, H) \log^3 n)$, and the single-logarithm bound when $p_{\mathrm{E}}(n, H) < 1/(3n)$. In [4, Theorems 1.4–1.5] they proved $N(H, F) \leq \mathbb{E}N(G(n, \min\{1, Lq\}), F)$ whenever all subgraphs of H have expected count at least one at density q, for F a clique, a cycle or a bounded-degree tree. The constant L is universal for cliques and cycles and depends on the degree bound for trees. Tran [16] obtained $q_f(\mathcal{F}_H) = O(p_{\mathrm{E}}(n, H) \log(2e(H)))$, which reduces the general loss to $O(\log^2(2e(H)))$. He also proved the second conjecture for trees and for graphs whose average degree is at least the logarithm of their maximum degree.

Method ancestry. The maximisation in Lemma 4.1 is the spread-link construction of Tran [16, Section 3, Lemma 3.1], with antecedents in [9, Claim 29] and [1, Lemma 2.4]. The copy-count identity (4.1) also appears in [11, Equation (5)]. The transition of Section 2 is the planted-set and posterior-resampling construction of Mossel, Niles-Weed, Sun and Zadik [10, Sections 2.1–2.3]. Tran [16, Section 4] proposed a coupling conjecture that would absorb successive conditional extensions while paying the logarithmic sampling cost once. The tree reduction of Section 3 achieves the required containment bound.

1.4 Conventions

log is the natural logarithm. For a finite set X and $0 \le \rho \le 1$, μ_ρ is the law of a random subset of X containing each element independently with probability ρ. Thus $\mu_\rho(y) = \rho^{|y|}(1-\rho)^{|X|-|y|}$. Three standard properties are used.

(P1) *Independence.* If $X_1 \cap X_2 = \varnothing$ and $W \sim \mu_\rho$, then any function of $W \cap X_1$ is independent of any function of $W \cap X_2$.

(P2) *Monotonicity.* If $\mathcal{E} \subseteq 2^X$ is closed under taking supersets and $\rho \le \rho'$, then $\mu_\rho(\mathcal{E}) \le \mu_{\rho'}(\mathcal{E})$. (Couple the two sets through independent uniform variables U_x, putting x in the set of density ρ when $U_x \le \rho$.)

(P3) *Unions.* The union of N independent sets of law μ_ρ has law μ_{ρ_*} with $\rho_* = 1-(1-\rho)^N$. Each element is missed by all of them with probability $(1-\rho)^N$, independently over elements.

Relation to the source. Theorem and lemma numbers agree with the source, and equation numbers do not. Facts A–D isolate steps that the source states in running text. Some symbols were renamed to avoid clashes. The outgoing level of a node (source: s) is j. The number of successes so far (source: t) is τ. The overlap size (source: R, r) is written $|A_c \cap A_b|$ with values g. In Section 4, $|S|$ and $|I|$ are written out (source: r, h_I). The names $K_W(c,b)$, Φ_j, λ_τ and ξ_τ are new names for objects the source describes in words. In preparing this rewrite every step of the proofs in Sections 2–4 was rechecked. None needed to be marked as unverified.

2 Resampling one spread family

This section is used only through Lemma 2.1 and Fact A. Both are applied at every node of a tree in Section 3.

2.1 Setting and the transition

Throughout this section X is a finite set and $(A_b)_{b\in B}$ is a finite indexed family of subsets of X ("labels"), with weights $\nu_b > 0$ and $\sum_b \nu_b = 1$. Distinct indices may carry the same label, and labels may be empty. For $a > 0$ the law ν is *a-spread* if

$$\mathbb{P}_{c\sim\nu}(J \subseteq A_c) = \sum_{b:\, J\subseteq A_b} \nu_b \le a^{|J|} \qquad (\varnothing \ne J \subseteq X). \tag{2.1}$$

If $a \le a'$, an a-spread law is a'-spread.

Fix $0 < \rho < 1$ and, for $y \subseteq X$, put

$$Z(y) = \sum_{d\in B} \nu_d\, \rho^{-|A_d|}\, \mathbf{1}\{A_d \subseteq y\}. \tag{2.2}$$

For a given set $W \subseteq X$ the *transition* is the following random experiment.

1. Choose a *source* $c \in B$ with law ν, and set $Y = W \cup A_c$.
2. Choose a *target* $b \in B$ with probability $\nu_b \rho^{-|A_b|}\mathbf{1}\{A_b \subseteq Y\}/Z(Y)$.
3. Record the *fragment* $T = A_b \setminus W$.

Step 2 makes sense because $Z(Y) \geq \nu_c \rho^{-|A_c|} > 0$. So for fixed W the pair (source, target) equals (c, b) with probability

$$K_W(c,b) = \nu_c \cdot \frac{\nu_b \, \rho^{-|A_b|} \, \mathbf{1}\{A_b \subseteq W \cup A_c\}}{Z(W \cup A_c)}, \qquad \text{and} \qquad \sum_{b \in B} K_W(c,b) = \nu_c. \tag{2.3}$$

In particular, *for every fixed W the source has law ν.* The target need not.

Intuition, not used later. If $W \sim \mu_\rho$, then Y is a random set with one label planted in it, and step 2 samples from the conditional law of the planted index given Y. The useful feature is that the target's label lies in $W \cup A_c$. So the part of A_b that W misses lies in $A_c \cap A_b$, and spread makes large overlaps of two labels unlikely.

Fact A (Elementary properties)**.** *Fix $W \subseteq X$.*

(a) If $K_W(c,b) > 0$ and $T = A_b \setminus W$, then

$$T \subseteq A_c \cap A_b, \qquad A_b \subseteq W \cup T. \tag{2.4}$$

(b) The numbers $K_W(c,b)$ and $Z(W \cup A_c)$ for $c, b \in B$, and the fragments $A_b \setminus W$, depend on W only through $W \cap \bigcup_{d \in B} A_d$.

Proof. (a) $K_W(c,b) > 0$ forces $A_b \subseteq W \cup A_c$, so $A_b \setminus W \subseteq A_c$. Trivially $T \subseteq A_b$ and $A_b \subseteq W \cup (A_b \setminus W)$. (b) $A_d \subseteq W \cup A_c$ holds if and only if $A_d \setminus A_c \subseteq W \cap A_d$. So every indicator in (2.2) and (2.3) is a function of $W \cap A_d$ for some d. Also $A_b \setminus W = A_b \setminus (W \cap A_b)$. □

Fact B (Law of the planted union)**.** *Let $W \sim \mu_\rho$ be independent of the source $c \sim \nu$, and $Y = W \cup A_c$. Then $\mathbb{P}(Y = y) = \mu_\rho(y)\, Z(y)$ for every $y \subseteq X$.*

Proof. Fix c. The event $W \cup A_c = y$ holds if and only if $A_c \subseteq y$ and $W \setminus A_c = y \setminus A_c$. The second condition prescribes the membership in W of each element of $X \setminus A_c$. When $A_c \subseteq y$ it has probability $\rho^{|y \setminus A_c|}(1-\rho)^{|X \setminus y|} = \mu_\rho(y) \rho^{-|A_c|}$. Multiply by ν_c and sum over c, using (2.2). □

2.2 The local lemma

Lemma 2.1 (Local resampling)**.** *Suppose that $|A_b| \leq m$ for all b, where m is a positive integer, that ν is a-spread, and that*

$$0 < \rho < 1, \qquad a/\rho \leq \mathrm{e}^{-50}.$$

Let $W \sim \mu_\rho$, independently of the source choice, and perform the transition. Call the outcome a local failure *if*

$$Z(Y) < \mathrm{e}^{-10m} \quad \text{or} \quad |T| > m/2.$$

Then

$$\mathbb{P}(\text{local failure}) \leq \mathrm{e}^{-9m}. \tag{2.5}$$

Moreover, for each fixed $b \in B$ and every function $f \geq 0$ of W that depends only on $W \cap (X \setminus A_b)$,

$$\mathbb{E}\Big[\mathbf{1}\{\text{target} = b,\ Z(Y) \geq \mathrm{e}^{-10m}\}\, f(W)\Big] \leq \mathrm{e}^{11m}\, \nu_b\, \mathbb{E} f(W). \tag{2.6}$$

The expectation on the right is over $W \sim \mu_\rho$.

Both conclusions are used in Step 4 of the proof of Lemma 3.2. (2.5) bounds the failure mass at a node. (2.6) is applied with f a quantity computed in the subtree below a child.

Proof. Plan. A small denominator is unlikely by Fact B. When the denominator is not small, $1/Z$ is at most e^{10m}, and the event $A_b \subseteq W \cup A_c$ costs $\rho^{|A_b \setminus A_c|}$. Each pair (c, b) then carries weight at most $\mathrm{e}^{10m}\nu_c\nu_b\rho^{-|A_c \cap A_b|}$. Spread bounds the sum of these weights. The triple (W, c, b) has $W \sim \mu_\rho$ and, given W, law K_W on pairs.

Step 1 (small denominator). By Fact B,

$$\mathbb{P}(Z(Y) < \mathrm{e}^{-10m}) = \sum_{y \subseteq X} \mu_\rho(y)\, Z(y)\, \mathbf{1}\{Z(y) < \mathrm{e}^{-10m}\} \le \mathrm{e}^{-10m} \sum_y \mu_\rho(y) = \mathrm{e}^{-10m}. \tag{2.7}$$

Step 2 (weight of a pair). Fix $c, b \in B$ and a function $\phi \ge 0$ of W. On the event $Z(W \cup A_c) \ge \mathrm{e}^{-10m}$ the denominator in (2.3) satisfies $1/Z \le \mathrm{e}^{10m}$. Also $A_b \subseteq W \cup A_c$ if and only if $A_b \setminus A_c \subseteq W$. Hence

$$\mathbb{E}\big[K_W(c, b)\, \mathbf{1}\{Z(W \cup A_c) \ge \mathrm{e}^{-10m}\}\, \phi(W)\big] \le \mathrm{e}^{10m}\nu_c\nu_b\, \rho^{-|A_b|}\, \mathbb{E}\big[\mathbf{1}\{A_b \setminus A_c \subseteq W\}\, \phi(W)\big]. \tag{2.8}$$

Two cases are needed. In both, $\mathbb{P}(A_b \setminus A_c \subseteq W) = \rho^{|A_b \setminus A_c|}$ and $|A_b| - |A_b \setminus A_c| = |A_c \cap A_b|$.

- For $\phi \equiv 1$ the right side of (2.8) equals
$$\mathrm{e}^{10m}\nu_c\nu_b\, \rho^{-|A_c \cap A_b|}. \tag{2.9}$$
- Let $\phi = f$ depend only on $W \cap (X \setminus A_b)$. The indicator depends only on $W \cap A_b$. By (P1) the two are independent, and the right side of (2.8) equals
$$\mathrm{e}^{10m}\nu_c\nu_b\, \rho^{-|A_c \cap A_b|}\, \mathbb{E}f(W). \tag{2.10}$$

Step 3 (overlaps are small). Fix b and, for an integer $g \ge 0$, let $\pi_b(g) = \sum_{c:\, |A_c \cap A_b| = g} \nu_c$. Then

$$\pi_b(g) \le \binom{m}{g} a^g \qquad (0 \le g \le m), \qquad \pi_b(g) = 0 \quad (g > m). \tag{2.11}$$

For $g = 0$ the bound is 1. For $g \ge 1$, if $|A_c \cap A_b| = g$ then $J = A_c \cap A_b$ is a g-element subset of A_b contained in A_c. By (2.1), $\pi_b(g) \le \sum_{J \subseteq A_b,\, |J| = g} \mathbb{P}_{c \sim \nu}(J \subseteq A_c) \le \binom{|A_b|}{g} a^g \le \binom{m}{g} a^g$. Consequently, for every set $\Gamma \subseteq \{0, \dots, m\}$,

$$\sum_{c:\, |A_c \cap A_b| \in \Gamma} \nu_c\, \rho^{-|A_c \cap A_b|} = \sum_{g \in \Gamma} \pi_b(g) \rho^{-g} \le \sum_{g \in \Gamma} \binom{m}{g} \Big(\frac{a}{\rho}\Big)^g. \tag{2.12}$$

Step 4 (large fragments). By Fact A(a), whenever $K_W(c, b) > 0$ the fragment satisfies $|T| \le |A_c \cap A_b|$. So $|T| > m/2$ forces $|A_c \cap A_b| > m/2$. Sum (2.8) with $\phi \equiv 1$ over such pairs and use (2.9), then (2.12) with $\Gamma = \{g : m/2 < g \le m\}$, then $\sum_b \nu_b = 1$:

$$\begin{aligned} \mathbb{P}(Z(Y) \ge \mathrm{e}^{-10m},\ |T| > m/2) &\le \mathrm{e}^{10m} \sum_b \nu_b \sum_{c:\, |A_c \cap A_b| > m/2} \nu_c\, \rho^{-|A_c \cap A_b|} \\ &\le \mathrm{e}^{10m} \sum_{m/2 < g \le m} \binom{m}{g} \Big(\frac{a}{\rho}\Big)^g \le \mathrm{e}^{10m}\, 2^m\, \mathrm{e}^{-25m}. \end{aligned} \tag{2.13}$$

The last step uses $(a/\rho)^g \le \mathrm{e}^{-50g} \le \mathrm{e}^{-25m}$ for $g > m/2$ and $\sum_g \binom{m}{g} = 2^m$.

Step 5 (proof of (2.5)). Add (2.7) and (2.13):

$$\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-10m} + 2^m \mathrm{e}^{-15m} = \mathrm{e}^{-9m}\big(\mathrm{e}^{-m} + (2\mathrm{e}^{-6})^m\big) \le \mathrm{e}^{-9m}.$$

For $m \ge 1$ the bracket is at most $\mathrm{e}^{-1} + 2\mathrm{e}^{-6} < 0.38$.

Step 6 (proof of (2.6)). The left side of (2.6) equals $\sum_c \mathbb{E}[K_W(c,b)\mathbf{1}\{Z(W\cup A_c) \ge \mathrm{e}^{-10m}\} f(W)]$. Apply (2.8) with $\phi = f$, then (2.10), then (2.12) with $\Gamma = \{0,\dots,m\}$:

$$\le \mathrm{e}^{10m}\nu_b\,\mathbb{E}f(W)\sum_c \nu_c\,\rho^{-|A_c\cap A_b|} \le \mathrm{e}^{10m}\nu_b\,\mathbb{E}f(W)\Big(1+\frac{a}{\rho}\Big)^m \le \mathrm{e}^{11m}\nu_b\,\mathbb{E}f(W).$$

The last step uses $(1+x)^m \le \mathrm{e}^{xm} \le \mathrm{e}^m$ for $0 \le x = a/\rho \le 1$. □

3 Covering a tree

3.1 Definitions and the tree theorem

A *probability tree on X* is a finite rooted tree with all leaves at the same depth k. Each non-leaf node v carries a probability distribution ν_v with positive weights on its set of children $\mathrm{Ch}(v)$. Each arc from v to a child b carries a label $A_v(b) \subseteq X$. Different arcs may carry equal labels, and nodes are never identified because labels agree. *Level j*, for $1 \le j \le k$, is the set of arcs from depth $j-1$ to depth j. It is the *outgoing level* of the nodes at depth $j-1$. The tree *has k levels*. A *path* always means a full root-to-leaf path P, and

$$\mathcal{U}(P) = \bigcup_{(v,b)\in P} A_v(b)$$

is the union of its labels. The single-node tree ($k=0$) has one path, with $\mathcal{U} = \varnothing$. It occurs only as the final state of an iteration.

The tree is *disjoint* if the labels along each path are pairwise disjoint. Level j has *capacity* m if all its labels have size at most m. Level j is *a-spread* if for every node v at depth $j-1$ the family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v is a-spread in the sense of (2.1). All requirements concern single child distributions. Nothing is assumed about the law of a full path.

Theorem 3.1 (Tree covering). *Let $\mathcal{T}$ be a disjoint probability tree on a finite set X with $k \ge 1$ levels. Suppose every level is σ-spread, where $\sigma > 0$, and that level i has capacity ℓ_i for positive integers $\ell_1,\dots,\ell_k$ satisfying*

$$\ell_{i+1} \ge 16\ell_i \qquad (1 \le i < k).$$

Set

$$r = \min\{1,\ 16\mathrm{e}^{50}\sigma(1+\log_2 \ell_k)\}.$$

Then

$$\mathbb{P}_{W\sim\mu_r}(\mathcal{U}(P) \subseteq W \text{ for some path } P \text{ of } \mathcal{T}) \ge \tfrac{2}{3}.$$

The theorem is applied once, in the proof of Theorem 1.1, to the tree of Lemma 4.1, with $\sigma = 128\,p_{\mathrm{E}}(n,H)$ and $\ell_k \le e(H)$. Its proof (Section 3.3) iterates the following lemma. The iteration replaces labels by smaller subsets, so empty labels and unequal label sizes must be allowed.

3.2 One simultaneous reduction

Lemma 3.2 (Reduction)**.** *Let $\mathcal{T}$ be a disjoint probability tree with $k \geq 1$ levels and positive integer capacities $m_1, \ldots, m_k$ satisfying $m_{j+1} \geq 16 m_j$. Suppose level j is α_j-spread, where*

$$0 < \alpha_j \leq a, \qquad \rho = \mathrm{e}^{50} a < 1.$$

There is a deterministic procedure which, given $W \sim \mu_\rho$, succeeds with probability at least $3/4$ and on success returns a disjoint probability tree $\mathcal{T}'$ with the same levels, such that:

(i) level j has capacity $\lfloor m_j/2 \rfloor$ and is $\alpha_j/(1-\mathrm{e}^{-m_j})$-spread;

(ii) for every path Q of $\mathcal{T}'$ there is a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q). \tag{3.1}$$

Proof. Plan. For each node of $\mathcal{T}$ and each W we compute the probabilities with which the transition of Section 2, run with that node's child law, fails or selects each child. A node is "good" if most of its mass goes to good children. The procedure succeeds when the root is good. Steps 1–5 show this has probability at least $\frac{3}{4}$. Step 6 builds $\mathcal{T}'$ from the good nodes and checks (i) and (ii). The tree $\mathcal{T}$ stays fixed in Steps 0–5. All quantities there are defined for every node of $\mathcal{T}$, whether or not it is later retained.

Step 0 (local masses). Let v be a non-leaf node with outgoing level j. Apply Section 2 to the family $(A_v(b))_{b \in \mathrm{Ch}(v)}$ with weights ν_v, the given ρ, and $m = m_j$. Write Z_v and K_W^v for the resulting quantities (2.2) and (2.3). The hypotheses of Lemma 2.1 hold. Labels have size at most m_j. The law ν_v is α_j-spread, hence a-spread. And $a/\rho = \mathrm{e}^{-50}$ with $0 < \rho < 1$. For fixed W and a pair (c, b) of children of v, *local failure* means $Z_v(W \cup A_v(c)) < \mathrm{e}^{-10 m_j}$ or $|A_v(b) \setminus W| > m_j/2$. Define

$$t_{vb}(W) = \sum_{c \in \mathrm{Ch}(v)} K_W^v(c,b)\, \mathbf{1}\{\text{no local failure at } (c,b)\}, \tag{3.2}$$

$$d_v(W) = \sum_{c,b \in \mathrm{Ch}(v)} K_W^v(c,b)\, \mathbf{1}\{\text{local failure at } (c,b)\}. \tag{3.3}$$

So $d_v(W)$ is the probability, at fixed W, of local failure at v, and $t_{vb}(W)$ that of non-failure with target b. Since K_W^v is a probability distribution on pairs,

$$d_v(W) + \sum_{b \in \mathrm{Ch}(v)} t_{vb}(W) = 1. \tag{3.4}$$

These numbers are explicit finite sums in W, and the procedure draws no further randomness. Sources and targets serve only to interpret the sums. Lemma 2.1 gives, for $W \sim \mu_\rho$,

$$\mathbb{E}\, d_v(W) \leq \mathrm{e}^{-9 m_j}, \tag{3.5}$$

and, for every $b \in \mathrm{Ch}(v)$ and every $f \geq 0$ depending only on $W \cap (X \setminus A_v(b))$,

$$\mathbb{E}\big[t_{vb}(W) f(W)\big] \leq \mathrm{e}^{11 m_j}\, \nu_v(b)\, \mathbb{E} f(W). \tag{3.6}$$

For (3.6): non-failure implies $Z_v \geq \mathrm{e}^{-10 m_j}$. So $t_{vb}(W) \leq \mathbb{P}(\text{target} = b,\ Z_v(Y) \geq \mathrm{e}^{-10 m_j} \mid W)$. Multiply by $f(W) \geq 0$, take expectations, and apply (2.6).

Step 1 (good nodes; definition of success). Fix W and put $\epsilon_j = \mathrm{e}^{-m_j}$. Declare all leaves good. Working upward, declare a node v with outgoing level j *good* if

$$\sum_{\substack{b\in\mathrm{Ch}(v)\\ b \text{ good}}} t_{vb}(W) \geq 1-\epsilon_j, \tag{3.7}$$

and *bad* otherwise. The procedure *succeeds* exactly when the root is good.

Step 2 (a potential that detects bad nodes). Set $D_v(W) = 0$ at leaves and, at a node with outgoing level j,

$$D_v(W) = \frac{d_v(W) + \sum_{b\in\mathrm{Ch}(v)} t_{vb}(W) D_b(W)}{\epsilon_j}. \tag{3.8}$$

By induction from the leaves, $D_v \geq 0$. *Claim: every bad node has $D_v(W) > 1$.* Leaves are good. Let v be bad and assume the claim for its bad children. Using $D_b \geq 0$ for good children and $D_b > 1$ for bad ones, the numerator of (3.8) is at least

$$d_v(W) + \sum_{b \text{ bad}} t_{vb}(W) = 1 - \sum_{b \text{ good}} t_{vb}(W) > \epsilon_j.$$

The equality is (3.4), and the inequality is the failure of (3.7).

Step 3 (locality: where disjointness is used). *Claim: for every non-leaf v and $b \in \mathrm{Ch}(v)$, the number $D_b(W)$ depends on W only through $W \cap (X \setminus A_v(b))$.* For a non-leaf node u, Fact A(b) shows that $d_u(W)$ and all $t_{uc}(W)$ depend only on $W \cap \bigcup_{c\in\mathrm{Ch}(u)} A_u(c)$. The local-failure indicator is a function of $Z_u(W \cup A_u(c))$ and $A_u(b) \setminus W$. By the recursion (3.8), $D_b(W)$ therefore depends only on W intersected with the union of the labels $A_u(c)$ over all arcs (u, c) in the subtree rooted at b. Each such arc lies on a root-to-leaf path through the arc (v, b): go from the root to v, then to b, down to u and c, and on to a leaf. Since $\mathcal{T}$ is disjoint, $A_u(c) \cap A_v(b) = \varnothing$. See Figure 1.

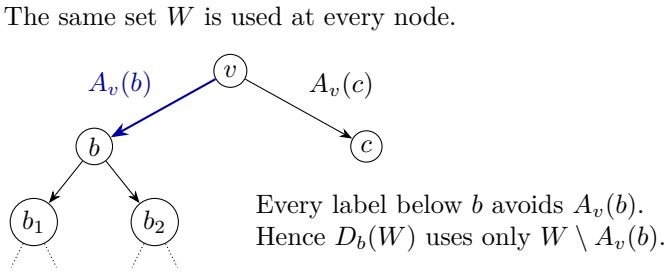


Figure 1: Every arc below b lies on a path through the arc into b, so D_b is a function of coordinates outside the incoming label. Labels on different branches (such as $A_v(c)$) may overlap $A_v(b)$.

Step 4 (one-level inequality). Now let $W \sim \mu_\rho$. For a node v with outgoing level j, take expectations in (3.8). Use (3.5). Use (3.6) with $f = D_b$, which is legitimate by Step 3 and $D_b \geq 0$. With $\epsilon_j^{-1} = \mathrm{e}^{m_j}$ this gives

$$\mathbb{E}D_v(W) \leq \mathrm{e}^{m_j}\Big(\mathrm{e}^{-9m_j} + \mathrm{e}^{11m_j} \sum_{b\in\mathrm{Ch}(v)} \nu_v(b)\, \mathbb{E}D_b(W)\Big) = \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j} \sum_{b\in\mathrm{Ch}(v)} \nu_v(b)\, \mathbb{E}D_b(W). \tag{3.9}$$

Step 5 (the root is bad with probability less than $\frac{1}{4}$). Define $\Phi_{k+1} = 0$ and $\Phi_j = \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j}\Phi_{j+1}$ for $j = k, k-1, \ldots, 1$. *Claim: $\mathbb{E}D_v(W) \leq \Phi_j$ for every node v at depth $j-1$.*

Leaves are at depth k and have $D = 0 = \Phi_{k+1}$. For a node at depth $j-1 \le k-1$ its children are at depth j. So (3.9), the induction hypothesis and $\sum_b \nu_v(b) = 1$ give the claim. Unrolling the recursion,

$$\Phi_1 = \sum_{i=1}^{k} \exp\Big(12\,(m_1 + \cdots + m_{i-1}) - 8m_i\Big). \tag{3.10}$$

Since $m_{i'} \le 16^{-(i-i')} m_i$ for $i' < i$, we get $m_1 + \cdots + m_{i-1} \le m_i \sum_{x \ge 1} 16^{-x} = m_i/15$. So the i-th exponent is at most $(\frac{12}{15} - 8)m_i \le -7m_i$. The m_i are strictly increasing positive integers, hence distinct. Therefore, for the root o, by Step 2 and Markov's inequality,

$$\mathbb{P}(o \text{ is bad}) \le \mathbb{P}(D_o(W) > 1) \le \mathbb{E}D_o(W) \le \Phi_1 \le \sum_{m \ge 1} \mathrm{e}^{-7m} = \frac{1}{\mathrm{e}^7 - 1} < \frac{1}{4}.$$

So the procedure succeeds with probability at least $\frac{3}{4}$.

Step 6 (the reduced tree). Fix W for which the root is good. Start from the root. At a good non-leaf node v with outgoing level j put

$$\gamma_v(W) = \sum_{b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j > 0.$$

Retain each good child b with $t_{vb}(W) > 0$. Give it weight $t_{vb}(W)/\gamma_v(W)$. Replace its incoming label by

$$T_b = A_v(b) \setminus W,$$

and continue recursively from b. Call the result $\mathcal{T}'$. All retained nodes are good. We verify the assertions.

(a) $\mathcal{T}'$ is a probability tree with the same levels. The weights at a retained non-leaf node are positive and sum to $\gamma_v/\gamma_v = 1$. Since $\gamma_v > 0$, at least one child is retained. So every retained node at depth less than k has a retained child, and all leaves of $\mathcal{T}'$ are at depth k.

(b) $\mathcal{T}'$ is disjoint. Every path of $\mathcal{T}'$ is a path of $\mathcal{T}$, with each label replaced by a subset.

(c) Capacity. If b is retained then $t_{vb}(W) > 0$. By (3.2) some pair (c, b) has no local failure. In particular $|T_b| = |A_v(b) \setminus W| \le m_j/2$, so $|T_b| \le \lfloor m_j/2 \rfloor$.

(d) Recovery (3.1). Let Q be a path of $\mathcal{T}'$ and P the same sequence of nodes viewed in $\mathcal{T}$. On each arc, $A_v(b) \subseteq W \cup T_b$. Taking unions gives $\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q)$.

(e) Spread. Let v be a retained non-leaf node with outgoing level j and let $J \subseteq X$ be nonempty. Under the new child law the probability that the label contains J is $\gamma_v(W)^{-1} \sum t_{vb}(W)$, summed over retained b with $J \subseteq T_b$. By (3.2), $t_{vb}(W) \le \sum_c K_W^v(c, b)$. If $K_W^v(c, b) > 0$ and $J \subseteq T_b$, then $J \subseteq T_b \subseteq A_v(c)$ by Fact A(a). Hence, using $\sum_b K_W^v(c, b) = \nu_v(c)$ from (2.3) and the α_j-spread of ν_v,

$$\sum_{b:\, J \subseteq T_b} t_{vb}(W) \le \sum_{c:\, J \subseteq A_v(c)} \sum_b K_W^v(c, b) = \sum_{c:\, J \subseteq A_v(c)} \nu_v(c) \le \alpha_j^{|J|}.$$

In words, the fragment lies inside the *source* label, and the source has law ν_v at every fixed W. Dividing by $\gamma_v \ge 1 - \epsilon_j$ and using $|J| \ge 1$,

$$\mathbb{P}_{\mathcal{T}'}(J \subseteq \text{label at } v) \le \frac{\alpha_j^{|J|}}{1 - \epsilon_j} \le \Big(\frac{\alpha_j}{1 - \mathrm{e}^{-m_j}}\Big)^{|J|}.$$

Items (a)–(e) give (i) and (ii). □

3.3 Iteration: proof of the tree theorem

Proof of Theorem 3.1. Plan. Apply Lemma 3.2 repeatedly with fresh random sets until every label is empty. Count how many sets are needed and take their union.

Step 0 (parameters). If $r = 1$ then $W = X$ and the conclusion is trivial. Suppose $r < 1$ and set

$$a = 4\sigma, \qquad \rho = \mathrm{e}^{50}a = 4\mathrm{e}^{50}\sigma, \qquad s = \lfloor \log_2 \ell_k \rfloor + 1.$$

Since $1 + \log_2 \ell_k \geq 1$, $r < 1$ gives $16\mathrm{e}^{50}\sigma < 1$. So $0 < \rho < \frac{1}{4}$. Also $2^{s-1} \leq \ell_k < 2^s$. For $\tau \geq 0$ and $1 \leq i \leq k$ define

$$m_i(\tau) = \left\lfloor \frac{\ell_i}{2^\tau} \right\rfloor, \qquad \alpha_i(\tau) = \sigma \prod_{\tau'=0}^{\tau-1} \left(1 - \mathrm{e}^{-m_i(\tau')}\right)^{-1}. \tag{3.11}$$

Two facts about floors are used. For real $x \geq 0$, $\lfloor \lfloor x \rfloor / 2 \rfloor = \lfloor x/2 \rfloor$. Indeed, if $y = \lfloor x/2 \rfloor$ then $2y \leq x < 2y + 2$, so $2y \leq \lfloor x \rfloor < 2y + 2$. Hence $\lfloor m_i(\tau)/2 \rfloor = m_i(\tau + 1)$. Also $\lfloor 16x \rfloor \geq 16 \lfloor x \rfloor$, so $m_{i+1}(\tau) \geq \lfloor 16\ell_i / 2^\tau \rfloor \geq 16\, m_i(\tau)$.

Step 1 (the procedure). Let $W_1, W_2, \ldots$ be independent sets of law μ_ρ. Maintain a current tree, initially $\mathcal{T}_0 = \mathcal{T}$. Suppose $\tau < s$ successes have occurred and the current tree is $\mathcal{T}_\tau$. Attempt number λ applies the procedure of Lemma 3.2 to $\mathcal{T}_\tau$ with the set W_λ. A failed attempt leaves the tree unchanged. A successful attempt returns a tree $\mathcal{T}'$, which is trimmed as follows. The levels i of $\mathcal{T}'$ with $m_i(\tau + 1) = 0$ form an initial segment, since $m_i(\tau + 1)$ is nondecreasing in i. By (i) all labels on them are empty. By a fixed rule, follow one path from the root of $\mathcal{T}'$ down through these levels to a node w. Let $\mathcal{T}_{\tau+1}$ be the subtree of $\mathcal{T}'$ rooted at w. If no level remains, $\mathcal{T}_{\tau+1}$ is a single node. Let λ_τ be the number of the τ-th successful attempt.

Step 2 (invariants). *Claim.* For $0 \leq \tau \leq s$, once τ successes have occurred:

(I1) $\mathcal{T}_\tau$ is a disjoint probability tree whose levels are, in order, the original levels i with $m_i(\tau) \geq 1$. Level i has capacity $m_i(\tau)$.

(I2) Level i of $\mathcal{T}_\tau$ is $\alpha_i(\tau)$-spread, and $\alpha_i(\tau) \leq 4\sigma = a$.

(I3) For every path Q of $\mathcal{T}_\tau$ there is a path P of $\mathcal{T}$ with $\mathcal{U}(P) \subseteq W_{\lambda_1} \cup \cdots \cup W_{\lambda_\tau} \cup \mathcal{U}(Q)$.

The bound $\alpha_i(\tau) \leq 4\sigma$. Let level i be present in $\mathcal{T}_\tau$, so $m_i(\tau) \geq 1$. The integers $x_{\tau'} = m_i(\tau')$ satisfy $x_{\tau'+1} = \lfloor x_{\tau'}/2 \rfloor < x_{\tau'}$ whenever $x_{\tau'} \geq 1$. So $x_0 > x_1 > \cdots > x_{\tau-1} \geq 2$ are distinct positive integers. The product in (3.11) is thus at most $\prod_{m \geq 1}(1 - \mathrm{e}^{-m})^{-1}$. This is less than 4. Using $-\log(1 - x) \leq x/(1 - x) \leq x/(1 - \mathrm{e}^{-1})$ for $0 \leq x \leq \mathrm{e}^{-1}$,

$$\sum_{m=1}^{\infty} -\log(1 - \mathrm{e}^{-m}) \leq \frac{1}{1 - \mathrm{e}^{-1}} \sum_{m=1}^{\infty} \mathrm{e}^{-m} = \frac{\mathrm{e}^{-1}}{(1 - \mathrm{e}^{-1})^2} < 0.93 < \log 4. \tag{3.12}$$

Induction. For $\tau = 0$, (I1)–(I3) are the hypotheses (with $P = Q$). Assume them for some $\tau < s$. By (I1), (I2) and Step 0, $\mathcal{T}_\tau$ satisfies the hypotheses of Lemma 3.2. The capacities $m_i(\tau)$ are positive integers with $m_{i+1}(\tau) \geq 16 m_i(\tau)$. The spread parameters satisfy $\alpha_i(\tau) \leq a$. And $\rho = \mathrm{e}^{50}a < 1$. $\mathcal{T}_\tau$ has at least one level because $m_k(\tau) \geq 1$ for $\tau \leq s - 1$. On success the lemma returns a disjoint tree $\mathcal{T}'$ whose level i has capacity $\lfloor m_i(\tau)/2 \rfloor = m_i(\tau + 1)$ and is $\alpha_i(\tau)/(1 - \mathrm{e}^{-m_i(\tau)}) = \alpha_i(\tau + 1)$-spread. Trimming preserves these. A subtree of a disjoint probability tree is one, capacity and spread are conditions on single nodes, and exactly the levels with $m_i(\tau + 1) \geq 1$ remain. This gives (I1) and (I2) for $\tau + 1$. For (I3), let Q' be a path of $\mathcal{T}_{\tau+1}$. Prefixing the trimmed path from

the root of $\mathcal{T}'$ to w, whose labels are empty, gives a path Q of $\mathcal{T}'$ with $\mathcal{U}(Q) = \mathcal{U}(Q')$. By (3.1) there is a path P' of $\mathcal{T}_\tau$ with $\mathcal{U}(P') \subseteq W_{\lambda_{\tau+1}} \cup \mathcal{U}(Q')$. By (I3) for τ there is a path P of $\mathcal{T}$ with $\mathcal{U}(P) \subseteq W_{\lambda_1} \cup \cdots \cup W_{\lambda_\tau} \cup \mathcal{U}(P')$. Combine the two inclusions.

Step 3 (after s successes a path is covered). Since $\ell_i \le \ell_k < 2^s$, $m_i(s) = 0$ for all i. By (I1), $\mathcal{T}_s$ is a single node. Its only path Q has $\mathcal{U}(Q) = \varnothing$. By (I3) there is a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq W_{\lambda_1} \cup \cdots \cup W_{\lambda_s}.$$

Step 4 ($4s$ attempts suffice with probability $\frac{2}{3}$). Before attempt λ the current tree is a function of $W_1, \dots, W_{\lambda-1}$. Conditionally on these, W_λ still has law μ_ρ. So by Lemma 3.2 and Step 2 the attempt succeeds with conditional probability at least $\frac{3}{4}$. Let $\xi_\tau = \lambda_\tau - \lambda_{\tau-1}$ (with $\lambda_0 = 0$) be the number of attempts used for the τ-th success. Decomposing according to the value of $\lambda_{\tau-1}$ and conditioning on the draws so far gives $\mathbb{P}(\xi_\tau > x+1) \le \frac{1}{4}\mathbb{P}(\xi_\tau > x)$. Hence $\mathbb{P}(\xi_\tau > x) \le 4^{-x}$ and $\mathbb{E}\xi_\tau = \sum_{x \ge 0} \mathbb{P}(\xi_\tau > x) \le \frac{4}{3}$. Thus $\mathbb{E}\lambda_s \le \frac{4}{3}s$, and by Markov's inequality

$$\mathbb{P}(\lambda_s > 4s) \le \frac{\mathbb{E}\lambda_s}{4s} \le \frac{1}{3}.$$

Step 5 (conclusion). Let $W_* = W_1 \cup \cdots \cup W_{4s}$. On the event $\lambda_s \le 4s$, which has probability at least $\frac{2}{3}$, Step 3 gives a path P with $\mathcal{U}(P) \subseteq W_*$. By (P3) and Bernoulli's inequality, W_* has law μ_{r_*} with

$$r_* = 1 - (1-\rho)^{4s} \le 4s\rho = 16\mathrm{e}^{50}\sigma s \le 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k) = r.$$

The event "$\mathcal{U}(P) \subseteq W$ for some path P" is closed under enlarging W. By (P2) its probability under μ_r is at least that under μ_{r_*}, hence at least $\frac{2}{3}$. □

Sanity check. For $k = 1$ the tree is a single σ-spread family of sets of size at most ℓ_1. The theorem then says that a random set of density $O(\sigma \log(2\ell_1))$ contains a member with probability at least $\frac{2}{3}$.

4 Application to graph containment

It remains to build, for the given H, a tree satisfying the hypotheses of Theorem 3.1.

4.1 Conventions for copies

Put $q = p_{\mathrm{E}}(n, H)$ and $\sigma = 128q$. By facts (b) and (c) of Section 1, $q > 0$ and every constraint in the definition of p_{E} holds at $p = q$.

Isolated vertices. Delete the isolated vertices of H. This does not change the event of containing H. A copy of the remaining graph in a graph on $[n]$ extends to a copy of the original H by assigning the deleted vertices to unused vertices of $[n]$. This is possible since H had at most n vertices and copies need not be induced. The number q remains the threshold of the *original* H.

Edge sets. From now on a graph without isolated vertices is identified with its edge set, and H also denotes its own edge set, so $|H| = h$. An edge set F has vertex set $V(F)$, the set of endpoints of its edges. Two edge sets are isomorphic if a bijection of vertex sets carries one onto the other. Let $X = \binom{[n]}{2}$. A *copy of F on $[n]$* is a subset of X isomorphic to F. For edge sets F and I let

$$M(F) = \#\{\text{copies of } F \text{ on } [n]\} = N(K_n, F), \qquad M(I, F) = \#\{F' \subseteq I : \ F' \cong F\} = N(I, F).$$

A subgraph isomorphic to a graph without isolated vertices is determined by its edge set, so these are the ordinary copy counts. (In terms of vertex maps, $M(F) = (n)_{v(F)}/|\operatorname{Aut}(F)|$ with $(n)_v = n(n-1)\cdots(n-v+1)$. This formula is not needed.) Set $M(\varnothing) = M(I, \varnothing) = 1$. Every edge subset $F \subseteq H$, on its vertex set, is a subgraph of the original H and has at most n vertices.

Fact C (Copy ratio)**.** *Let $F \subseteq I$ be edge sets with $|V(I)| \le n$. Let $\boldsymbol{I}$ be a uniformly random copy of I on $[n]$ and F_0 any fixed copy of F on $[n]$. Then*

$$\mathbb{P}(F_0 \subseteq \boldsymbol{I}) = \frac{M(I,F)}{M(F)}. \tag{4.1}$$

Proof. A permutation of $[n]$ maps the set of copies of I bijectively to itself, so it preserves the law of $\boldsymbol{I}$. Any copy of F can be mapped to any other by a permutation of $[n]$. Hence $\mathbb{P}(F_0 \subseteq \boldsymbol{I})$ is the same for all $M(F)$ copies F_0. Summing over them gives the expected number of copies of F contained in $\boldsymbol{I}$. This number is $M(I,F)$, because $\boldsymbol{I} \cong I$. □

Fact D (First moments at q)**.** *For every nonempty edge subset $F \subseteq H$,*

$$M(F)\, q^{|F|} \ge \tfrac{1}{2}. \tag{4.2}$$

Proof. F is a subgraph of the original H. So its constraint $\mathbb{E}N(G(n,q),F) \ge \frac{1}{2}$ holds at $q = p_{\mathrm{E}}(n,H)$. By (1.1) the left side is $M(F)q^{|F|}$. □

4.2 The hierarchy

Lemma 4.1 (Graph hierarchy)**.** *There is a disjoint probability tree on $X = \binom{[n]}{2}$ whose path unions are copies of H with its isolated vertices removed, such that every level is σ-spread and its label sizes $\ell_1, \dots, \ell_k$ satisfy*

$$1 \le \ell_1, \qquad \ell_{i+1} \ge 16\ell_i, \qquad \ell_k \le e(H).$$

Here $\sigma = 128q$, all labels on level i have size exactly ℓ_i, and $k \ge 1$. The lemma is used once, in the proof of Theorem 1.1.

Proof. Plan. Peel H down to $\varnothing$ by repeatedly passing to a subset that maximises a normalised containment probability. Fact D makes each step shrink the edge set by a factor of more than 17. Maximality is exactly the spread condition for the extension back up.

Step 1 (a maximiser is small). For a nonempty edge set $I \subseteq H$ and $S \subseteq I$ define

$$R_\sigma(I,S) = \sigma^{-|S|}\, \mathbb{P}(S_0 \subseteq \boldsymbol{I}), \tag{4.3}$$

where $\boldsymbol{I}$ is a uniform copy of I on $[n]$ and S_0 is any fixed copy of S. By Fact C this does not depend on S_0. Note $R_\sigma(I,\varnothing) = 1$. Let $S \ne \varnothing$. By (4.1), (4.2) for $F = S$, and $M(I,S) \le \binom{|I|}{|S|}$ (copies of S in I are $|S|$-element subsets of I),

$$R_\sigma(I,S) = \sigma^{-|S|}\frac{M(I,S)}{M(S)} \le \sigma^{-|S|}\binom{|I|}{|S|}\, 2q^{|S|} = 2\binom{|I|}{|S|}128^{-|S|}.$$

If moreover $|S| \ge |I|/17$, then $\binom{|I|}{|S|} \le (\mathrm{e}|I|/|S|)^{|S|} \le (17\mathrm{e})^{|S|}$ gives

$$R_\sigma(I,S) \le 2\left(\frac{17\mathrm{e}}{128}\right)^{|S|} < 1, \tag{4.4}$$

because $17\mathrm{e}/128 < 0.37$ and $|S| \geq 1$. Since the empty set has value 1, every maximiser S of $R_\sigma(I, \cdot)$ over subsets of I satisfies $|S| < |I|/17$. In particular $S \neq I$.

Step 2 (the chain). Start with $I = H$. Choose a maximiser S of $R_\sigma(I, \cdot)$ by a fixed tie-breaking rule. Replace I by S and repeat until $S = \varnothing$. Sizes strictly decrease, so this terminates. In reverse order the sets form a chain

$$\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = H, \qquad |H_{i-1}| < |H_i|/17, \qquad k \geq 1,$$

in which H_{i-1} maximises $R_\sigma(H_i, \cdot)$ over all edge subsets of H_i.

Step 3 (the tree). The root is $\varnothing$, the unique copy of H_0. A node at depth $i-1$ is a copy S_0 of H_{i-1} on $[n]$ together with its history. Distinct histories are distinct nodes. Its children are all copies I_0 of H_i on $[n]$ with $I_0 \supseteq S_0$, with the uniform distribution. The arc to I_0 has label $I_0 \setminus S_0$. Children exist. Indeed, S_0 is the image of H_{i-1} under an injection $V(H_{i-1}) \to [n]$. This extends to an injection $V(H_i) \to [n]$ because $|V(H_i)| \leq n$, and the image of H_i is a copy containing S_0. So this is a probability tree with k levels.

Step 4 (disjointness and sizes). Along a path the copies are nested, $\varnothing \subset I_1 \subset \cdots \subset I_k$, and the labels are the successive differences. They are pairwise disjoint with union I_k, a copy of H. Every label on level i has size $\ell_i = |H_i| - |H_{i-1}| \geq 1$. Moreover

$$\ell_{i+1} = |H_{i+1}| - |H_i| > 17|H_i| - |H_i| = 16|H_i| \geq 16\ell_i, \qquad \ell_k \leq |H_k| = e(H).$$

Step 5 (spread). Fix a node S_0 at depth $i-1$ and a nonempty $J \subseteq X$. Let A be the random label of the arc to its child. The child is uniform among the copies of H_i containing S_0. That is, it has the law of a uniform copy $\boldsymbol{I}$ of H_i conditioned on $S_0 \subseteq \boldsymbol{I}$, and $A = \boldsymbol{I} \setminus S_0$. If $J \cap S_0 \neq \varnothing$ then $\mathbb{P}(J \subseteq A) = 0$. Otherwise

$$\mathbb{P}(J \subseteq A) = \frac{\mathbb{P}(S_0 \cup J \subseteq \boldsymbol{I})}{\mathbb{P}(S_0 \subseteq \boldsymbol{I})}. \tag{4.5}$$

If the numerator is zero there is nothing to prove. If it is positive, $S_0 \cup J$ lies in some copy of H_i. Pulling back along the isomorphism, $S_0 \cup J$ is a copy of an edge subset $U \subseteq H_i$ with $|U| = |S_0| + |J|$. Also S_0 is a copy of H_{i-1}. Maximality of H_{i-1} gives $R_\sigma(H_i, U) \leq R_\sigma(H_i, H_{i-1})$, that is,

$$\sigma^{-(|S_0|+|J|)}\, \mathbb{P}(S_0 \cup J \subseteq \boldsymbol{I}) \leq \sigma^{-|S_0|}\, \mathbb{P}(S_0 \subseteq \boldsymbol{I}).$$

With (4.5) this is $\mathbb{P}(J \subseteq A) \leq \sigma^{|J|}$. So every level is σ-spread. □

Sanity check. If H is a single edge then $q = 1/(2\binom{n}{2})$, $k = 1$, and the root's children are the $\binom{n}{2}$ edges with uniform weights. Spread requires $1/\binom{n}{2} \leq 128q$, and indeed $1/\binom{n}{2} = 2q$.

4.3 Proof of the main theorem

Proof of Theorem 1.1. The tree of Lemma 4.1 is a disjoint probability tree on $X = \binom{[n]}{2}$ with $k \geq 1$ levels. Every level is σ-spread with $\sigma = 128q > 0$. Level i has capacity ℓ_i, a positive integer, and $\ell_{i+1} \geq 16\ell_i$. Theorem 3.1 applies. So a random edge set of density $\min\{1, 16\mathrm{e}^{50} \cdot 128q\,(1 + \log_2 \ell_k)\}$ contains $\mathcal{U}(P)$ for some path P with probability at least $\frac{2}{3}$. This $\mathcal{U}(P)$ is a copy of H without its isolated vertices. Since $\ell_k \leq h$ and $16 \cdot 128 = 2048$, this density is at most

$$r = \min\{1,\ 2048\mathrm{e}^{50} q(1 + \log_2 h)\}.$$

By (P2) the same holds at density r. A random edge set of density r on X is $G(n,r)$. By the remark on isolated vertices, $G(n,r)$ contains a copy of the original H with probability at least $\frac{2}{3} \geq \frac{1}{2}$. Hence $p_c(n,H) \leq r$, which is (1.2).

Finally, $h \leq \binom{n}{2} \leq n^2$ and $\log_2 n \geq 1$ give $1 + \log_2 h \leq 1 + 2\log_2 n \leq 3\log_2 n$. So $p_c(n,H) \leq 3 \cdot 2048\, e^{50} q \log_2 n = 6144\, e^{50} p_E(n,H) \log_2 n$. □

5 Verification checklist

These are the steps on which the argument rests, in suggested order of checking.

1. **Locality of the potential** (Lemma 3.2, Step 3). $D_b(W)$ must be a function of W outside the incoming label $A_v(b)$. This is the only place where disjointness along paths is used for the probability estimate. Check that d_u and t_{uc} are defined from the *fixed input tree* for all nodes, not from the pruned tree, and that they inspect W only on the child labels of u (Fact A(b)).

2. **Distortion bound** (2.6) (Lemma 2.1, Steps 2 and 6). The independence in (2.10) needs f to depend only on $W \setminus A_b$ and the indicator only on $W \cap A_b$. The exponent bookkeeping is $\rho^{-|A_b|}\rho^{|A_b \setminus A_c|} = \rho^{-|A_c \cap A_b|}$.

3. **Exponent arithmetic in the potential** (Lemma 3.2, Steps 4–5). $e^m e^{-9m} = e^{-8m}$ and $e^m e^{11m} = e^{12m}$. Then $12 \cdot \frac{1}{15} - 8 \leq -7$ uses $m_{j+1} \geq 16 m_j$, and distinctness of the m_j gives the bound $\sum_{m \geq 1} e^{-7m} < \frac{1}{4}$.

4. **Failure bound** (2.5) (Lemma 2.1, Steps 1, 4, 5). The density identity of Fact B, the implication $|T| > m/2 \Rightarrow |A_c \cap A_b| > m/2$ from Fact A(a), and $e^{-10m} + 2^m e^{-15m} \leq e^{-9m}$.

5. **Spread of the reduced tree** (Lemma 3.2, Step 6(e)). The fragment is compared with the source, whose law is ν_v at every fixed W. The only loss is the normalisation $\gamma_v \geq 1 - e^{-m_j}$, which needs the retained node to be good.

6. **Accumulated spread loss and attempt count** (Theorem 3.1, Steps 2 and 4). The capacities $\lfloor \ell_i / 2^{\tau'} \rfloor$ entering the product are distinct positive integers and $\prod_{m \geq 1}(1 - e^{-m})^{-1} < 4$ (3.12). Each attempt uses a fresh set, so the $\frac{3}{4}$ bound holds conditionally on the past.

7. **The hierarchy** (Lemma 4.1, Steps 1 and 5). Fact D is applied to S as a subgraph of the original H with ordinary copy counts. $2(17e/128)^{|S|} < 1$ forces $|S| < |I|/17$. Maximality is over *all* edge subsets of H_i, so it covers the set $U \cong S_0 \cup J$.

8. **Constants** (proofs of Theorems 3.1 and 1.1). $\rho = 4e^{50}\sigma$, $4s$ draws with $s \leq 1 + \log_2 \ell_k$, and $\sigma = 128q$, giving $4 \cdot 4 \cdot 128 = 2048$ and $3 \cdot 2048 = 6144$.

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